1. Machine Learning

- Life without ML
- Life with ML
- What is ML
- Types of ML
- The right ML solutions
- ML Algorithms.

Learning - ① Bait Shyness - Rats learning to avoid poisonous baits

Learning is the process of converting experience into expertise or knowledge.

② Pigeon Superstition - To know best behaviour at diff. time

- Life without machine learning

In our daily life, we search our data/information on Google for any random search. It collect all the information and represents relevant search.

Facial Recognition - ML made it possible for FB or Instagram, now you can identify/search any people you want to tag. It saves more and more time.

Voice Recognition - Siri, Iris, also used for making your life easy by using ML

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In the real world, we are surrounded by humans who can learn everything from their experiences with their learning capacity, and we have computers & machines which work on our instructions, But can a machine also learn from experiences or past data like a human does? So this is the role of machine learning.

* Machine learning is said as a subset of artificial Intelligence, mainly concerned with the development of algorithms which allow a computer to learn from the data & past experiences on their own.

→ The term machine learning was introduced by Arthur Samuel in 1959. So it can summarized as -

"Machine learning enables a machine to automatically learn from data, improve performance from experiences, and predict things without being explicitly programmed".

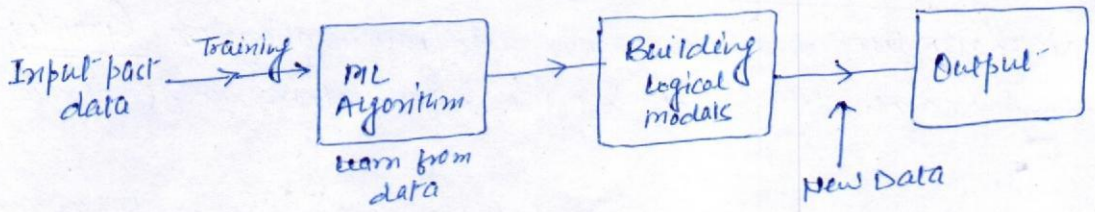
How it works? With the help of sample historical data, which is known as training data, machine learning algorithms build a mathematical model that helps in making predictions or decisions without being explicitly programmed.

So. we can say,

A machine learning system learns from historical data, builds the prediction models, and whenever it recieves new data, predicts the output for it.

The accuracy of predicted output depends upon the amount of data, as the huge amount of data helps to build a better model which predicts the o/p more accurately.

Block diagram for working of ML



→ features of machine learning:

- Machine learning uses data to detect various patterns in a given dataset.
- It can learn from past data and improve automatically.
- It is a data driven technology.
- Machine learning is much similar to data mining as it also deals with huge amount of the data.

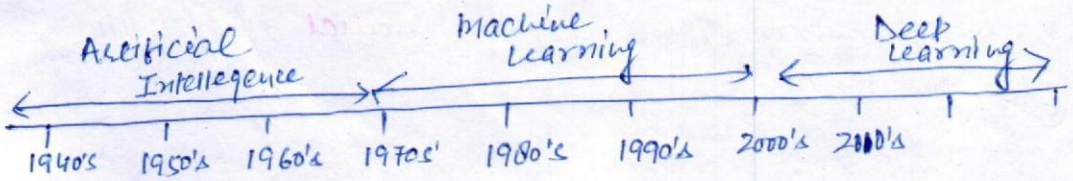
→ Importance & uses:-

Currently ML is used in Self-driving cars, Cyber fraud detection, face recognition & freind suggestion by facebook etc.

History of machine learning:-

Before some years (about 40-50 yrs), ML was science fiction, but today it is the part of our daily life.

ML is making our life day to day easy from self-driving cars to Amazon Virtual assistant "Alexa".



The early history of ML (pre 1940) -

- 1834: In 1834, Charles Babbage, the father of the Computer, Concieved a device that could be programmed with punch cards, however the machine was never built, but all modern computers rely on its logical structures.

- 1936: In 1936, Alan Turing gave a theory that how a machine can determine and execute a set of instructions

The era of stored program computers:

- 1940 - In 1940, the first manually operated computer, "ENIAC" was invented, which was the first electronic general-purpose comp. After that stored prog. comp. such as "EDSAC" in 1949 and "EDVAC" in 1951 were invented.

- 1943 - In 1943, a human neural network was modeled with an electrical circuit. In 1950, the scientists started applying their idea to work and analyzed how human neurons might work.

Computer Machinery and Intelligence:

- 1950 : In 1950, Alan Turing published a paper "Computer Machinery and Intelligence", on the topic of artificial Intelligence. In this paper, he asked, "Can machines think?"

Machine Intelligence in Games:-

- 1952 - Arthur Samuel, who was the pioneer of ML, created a prog. that helped an IBM computer to play a checker game. It performed better more it played.

- 1959 - In 1959, the term "Machine Learning" was first coined by "Arthur Samuel"

The first AI Winter: -

⇒ The duration of 1974 to 1980 was the tough time for AI & ML researchers, and this duration was called as "AI Winter".

In this duration, failure of machine translation occurred, and people had reduced their interest from AI, which led to reduced funding by gov. to researchers.

Machine learning from theory to reality:-

- 1959:- In 1959, the first neural network was applied to a real-world problem to remove echoes over phone lines using an adaptive filter.

- 1985:- In 1985, Terry Sejnowski and Charles Rosenberg invented a neural network NETtalk, which was able to teach itself how to correctly pronounce 20,000 words in one week.

- 1997:- The IBM's Deep Blue intelligent computer won the chess game against the chess expert Garry Kasparov, and it became the first computer which had beaten a human chess expert.

Machine Learning at 21 Century

- 2006:- In the year 2006, computer scientist Geoffrey Hinton has given a new name to neural net research as "deep learning" and nowadays, it has become one of the most trending tech.

- 2012:- In 2012, Google created a deep neural network which learned to recognize the image of humans and cats in YouTube videos.

- 2014:- In 2014, the Chabot "Eugen Goostman" cleared the Turing Test. It was the first Chabot who convinced the 33% of human judges that is was not a machine.

- 2014:- Deepface was deep neural network created by facebook, and they claimed that it could recognize a person with the same precision as a human can do.

- 2016:- AlphaGo beat the world's number second Lee Sedol at Go game. In 2007 it beat the no. one player of this game Ke Jie.

- 2017:- In 2017, the Alphabet's jigsaw team built an ~~integ~~ intelligent system that was able to learn the online trolling. It used to read millions of comments of diff. websites to learn to stop online trolling.

Machine Learning at Present:- Now machine learning has got a great advancement in its research, and it is present everywhere around us, such as self driving cars, Amazon Alexa, Catboats, reccomender system and many more.

It includes Supervised, Unsupervised, and reinforce learning with clustering, classification, decision tree, SVM algorithms etc.

Modern ML models → various predictions, including weather prediction, disease predition, stock market analysis etc.

Applications of Machine Learning:

Machine learning is a buzzword for today's technology, and it is growing very rapidly day by day.

We are using ML in our daily life even without knowing it such as Google maps, Google assistant, Alexa etc.

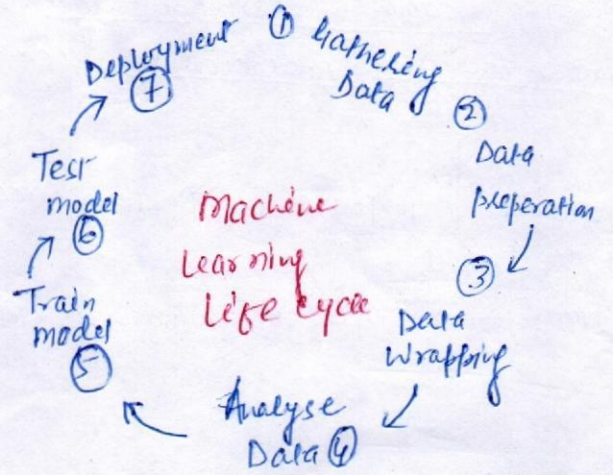
Some most ~~trend~~ trending real world appⁿ of ML -

- Automatic Language Translation – GNMT (Google Neural machine Translation) translates text into our familiar lang.
- Medical Diagnosis – To build 3D models for brain examine, brain tumors or other brain deseases.
- Stock market Trading – for up & downs share, long short term mem. neural network is used for prediction of stock market
- Online fraud Detection – Online transactions, to find fake ids, fake accounts to detect this feed forward Neural N/W help us to check transaction.
- Virtual Personal Assistant – Google Assistant, Alexa, cortana, Siri help us in finding information using our voice instruction.
- Email spam & malware filtering – Multi-Layer Perception, Decision Tree, Naive Bayes classifier Algos are used for email spam & malware filtering.
- Self Driving Cars – Tesla, car company using unsupervised learning to train the car model to detect people & objects.
- Product recommendations – Online Shopping, Amazon, Netflix etc.
- Traffic Prediction – { Real Time Location, Average time has taken } It takes information from the user & sends back to its db to improve the performance
- Speech Recognition – search by voice (Google) / Google assistant, Siri, Cortana, Alexa etc.
- Image " " – Face Detection → Automatic friend tagging suggection (FB)

Life Cycle of Machine Learning:-

Machine learning Life cycle is a cyclic process to build an efficient machine learning project. The main purpose of the life cycle is to find a solⁿ to the problem or project.

ML life cycle involves 7 major steps –



In the complete life cycle process, to solve a problem, we create a ML system called "Model" and this model is created by providing "training".

① Gathering Data:

Data Gathering is the first step of the ML life cycle. The goal of this step is to identify and obtain all data-related problems.

The data can be collected from diff. sources such as files, database, internet or mobile devices.

The quantity and quality of the collected data will determine the efficiency of the output.

② Data Preparation:

After collecting the data, we need to prepare it for further steps. In this step we put our data into a suitable place & prepare it to use in our machine learning training.

③ Data Wrangling:- is the process of cleaning and converting raw data into a usable format.

it is the process of cleaning the data, selecting the variable to use, and transforming the data in a proper format.

It is not necessary that collected data is usable always as, in real world data may have various issues, including

- missing values,
- Duplicacy
- Invalid data
- Noise

So we use various filtering techniques to clean the data.

④ Data Analysis:

Cleaned and prepared data is passed on to the analysis step. This step involves:-

- Selection of analytical techniques
- Building models
- Review the result.

This step starts with the determination of the type of the problems, where we select the machine learning tech. such as Classification, Regression, Cluster Analysis, Association etc.

⑤ Train Model-

We train our model to improve its performance for better outcome of the problem.

We use datasets to train the model using various ML algos.

6. Test Model :- Once our machine learning model has been trained on a given dataset, we test the model.
In this step, we check for the accuracy of our model by providing a test dataset to it.
"Testing the model determines the percentage accuracy of the model as per the requirement of project or problem.

7. Deployment :- We deploy the model in the real world system.
Before deploying the project, we will check whether it is improving its performance using available data or not.

Diff. b/w AI & ML :-

AI and ML are the part of computer science that are correlated with each other. These 2 tech. are used for creating intelligent systems.

On a broad level, we can differentiate both AI and ML as:-

"AI is a bigger concept to create intelligent machines that can simulate human thinking capability and behaviour, whereas, ML is an appⁿ or subset of AI that allows machines to learn from data without being programmed explicitly."

Some differences are :-

AI	ML
1. The goal of AI is to make a smart comp system like humans to solve complex problems.	1. The goal of ML is to allow machines to learn from data so that they can give accurate results/output.
2. In AI, we make intelligent systems to perform any task like a human	2. we teach machines with data to perform a particular task and give an accurate result.
3. AI has a very wide range of scope.	3. machine learning has a limited scope.
4. AI is working to create an intelligent system which can perform various complex tasks.	4. ML is working to create machines that can perform only those specific tasks for which they are trained.
5. It includes learning, reasoning, and self correction	5. It includes learning and self correction when introduced with new data.
6. AI completely deals with structured, semi structured and unstructured data.	6. ML deals with structured and semi structured data.
7. AI can be - weak AI, general AI and strong AI	7. ML → supervised, unsupervised & reinforcement learning.